

Technical Documentation

I experimented with a generative spider web system created in Python using L-System rewriting, procedural drawing, and ColabTurtle. The system begins with an axiom and repeatedly applies replacement rules for a specified number of iterations. The generated string contains symbols such as F, +, -, [, and], which represent drawing or structural instructions. I combined the L-System-generated structure with procedural geometry because this allowed the outputs to remain recognizable as spider webs while still being controlled by generative rules.

I created three web patterns using different L-System rules. The Orb Web uses $X \rightarrow F+X$, where repeated F commands create radial support threads and + changes the direction of growth. The Corner Web uses $X \rightarrow F+FX$. The number of forward commands generated by this rule influences the density of the support structure. The Damaged Web uses the branching rule $X \rightarrow F[+X][-X]$. Its generated structure determines the complexity of the support system, while controlled randomness changes thread angles, lengths, and connections.

The generator also includes adjustable parameters for iterations, web layers, size, color, line thickness, and random variation. Instead of manually positioning each drawing, the program calculates its position and safe drawing radius based on the canvas dimensions. This makes the system more flexible and prevents generated points from leaving the ColabTurtle canvas.

One challenge during development was that my first generated patterns resembled snowflakes or repeated polygons more than spider webs. I solved this by separating the web into two structural parts: radial support threads and connecting capture threads. Another problem occurred when generated coordinates became negative, which ColabTurtle does not allow. I solved this by calculating the center and available drawing space automatically from the canvas dimensions. These changes made the generator more reliable while maintaining the visual characteristics of a spider web.

Creative Statement

I wanted to explore the relationship between repetition, structure, and imperfection in spider webs. I chose spider webs because they naturally resemble rule-based systems. They contain repeating radial and connecting threads, but real webs are rarely perfectly symmetrical. Their appearance changes depending on where they are constructed, their surrounding environment, and whether parts of the web have been damaged.

I began by trying to create a recognizable circular orb web. My early experiments often looked more like snowflakes because the repeated lines were too symmetrical and did not contain enough connecting structure. I eventually separated the pattern into support threads and connecting threads, which made the result more recognizable as a web. From there, I experimented with how the same idea could create different structures. The Corner Web represents a web adapting to an architectural space, while the Damaged Web explores irregularity by changing thread lengths and angles and removing some connections.